

# Two Different Objects: The DiffusionPDE Guided-ODE Baseline is Not the Girsanov-Corrected SMC Target

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## Abstract

This repository contains two sampling procedures that are often compared as if they estimated the same quantity. The first is the guided probability-flow ODE (PF-ODE) of Huang et al. (2024) — a deterministic denoising trajectory with a post-hoc loss-gradient correction — which is the baseline in `scripts/generate_*.py`. The second is the Girsanov-corrected sequential Monte Carlo (SMC) construction of the companion note (Denoising-SMC notes, 2026a), implemented on Burgers in `scripts/generate_burgers_gem.py`. This note makes the relationship between the two precise and shows that they are not different discretizations of a common target: the SMC framework targets the posterior  $p(x_0 | y) \propto p(y | x_0)p(x_0)$ , whereas the baseline is a deterministic map whose implicit target, when read as a guided PF-ODE, is a schedule-dependent cold reconstruction rather than the posterior, obtained by adding a constant multiple of the loss gradient to the state after each ODE step. We give (i) the principled guided PF-ODE that a twist induces, and show the schedule-dependent coefficient the baseline omits; (ii) an exact side-by-side of the update equations; (iii) a three-way disambiguation of the baseline, the deterministic limit of the guided SDE, and the full SMC sampler; and (iv) two implementation-level caveats that matter if the baseline convention is imported into the SMC weight: flat guidance invalidates the closed-form Gaussian kernel ratio, and the PDE-residual term used as the twist is not a likelihood. The note reports the modeling issues; it makes no code changes and takes no position on which method is preferable for the baseline’s own reconstruction goal.

## 1 Introduction and scope

The upstream code of Huang et al. (2024), retained here, solves forward and inverse PDE problems under partial observation by denoising a Gaussian field with a probability-flow ODE and, at each step, subtracting gradients of a sparse-observation loss and a PDE-residual loss (Huang et al., 2024, Algorithm 1). The paper describes this as guided diffusion and, via a diffusion-posterior-sampling (DPS) approximation (Chung et al., 2023), writes the guiding term as a correction to the score (Huang et al., 2024, Eq. (8)). The SMC extension in this repository (Denoising-SMC notes, 2026a,b) instead treats guided diffusion as *importance sampling on path space*, with a Girsanov/twisted correction that makes the sampler consistent for the posterior, and a Hessian-free potential representation (Denoising-SMC notes, 2026b,c). The two are routinely placed on the same axis and scored with the same metric (relative  $L^2$  error to the ground-truth field), which suggests they approximate the same object. They do not.

The purpose of this note is to state precisely what each construction computes and where the frameworks genuinely diverge. We restrict attention to what is actually written in the released

code; where the paper and the code differ, we say so, since that difference is itself one of the mechanisms by which the baseline departs from any posterior interpretation. We emphasize at the outset that the baseline is a *reconstruction* procedure and the paper frames it as such (“fill in the missing information and solve a PDE by modeling the joint distribution” Huang et al., 2024); the observation that it is not a posterior sampler is not a criticism of its stated goal, only a statement that it cannot be used to validate an SMC construction of that posterior.

**Notation and sources.** We use  $\sigma$  for the variance-exploding noise level,  $D_\theta$  for the pretrained denoiser,  $L$  for a terminal likelihood, and  $\ell$  for a log-twist. File references are to paths in this repository.

## 2 The Girsanov-corrected SMC framework

We recall the construction of Denoising-SMC notes (2026a), which is the target the SMC code is built to estimate.

### 2.1 Target and reverse processes

Let  $x_0 \in \mathbb{R}^d$  be the quantity of interest and  $y$  the observation, with likelihood  $p(y \mid x_0)$  and a diffusion-model prior  $p(x_0)$ . The posterior target is

$$p(x_0 \mid y) \propto p(y \mid x_0)p(x_0). \quad (1)$$

With forward noising time  $t \in [0, T]$  and reverse (sampling) time  $\tau := T - t$ , define the reverse state  $\xi_\tau := x_{T-\tau}$ , so that  $\xi_0 = x_T$  is pure noise and  $\xi_T = x_0$  is data. The unguided reverse process has path measure  $\mathbb{P}$  and dynamics

$$d\xi_\tau = a(\xi_\tau, \tau) d\tau + g(\tau) dW_\tau^\mathbb{P}, \quad \Sigma(\tau) := g(\tau)g(\tau)^\top. \quad (2)$$

Fix a positive, smooth *twist*  $\phi_\tau(x) > 0$  and put

$$\ell_\tau(x) := \log \phi_\tau(x), \quad b_\tau(x) := \nabla_x \ell_\tau(x), \quad H_\tau(x) := \nabla_x^2 \ell_\tau(x). \quad (3)$$

The guided proposal, with path measure  $\mathbb{Q}^\phi$ , augments the drift by  $\Sigma b$ :

$$d\xi_\tau = [a(\xi_\tau, \tau) + \Sigma(\tau) b_\tau(\xi_\tau)] d\tau + g(\tau) dW_\tau^\mathbb{Q}. \quad (4)$$

The path-space target is  $p(y \mid \xi_T) \mathbb{P}(d\xi_{0:T})$ , so the path weight is

$$W(\xi_{0:T}) = p(y \mid \xi_T) \frac{d\mathbb{P}}{d\mathbb{Q}^\phi}(\xi_{0:T}). \quad (5)$$

If  $\phi_\tau$  equals the exact intermediate likelihood  $p(y \mid \xi_\tau) = \int p(y \mid \xi_T)p(\xi_T \mid \xi_\tau)d\xi_T$ , then the guided drift is  $\mathbb{P}$ -harmonic, (4) is the Doob  $h$ -transform of (2), and its terminal marginal is exactly (1). A general surrogate  $\phi$  makes the proposal suboptimal but leaves the *validity* of (5) untouched: the surrogate affects only weight variance, never the target, provided the terminal correction below is retained or the surrogate is terminally consistent.

## 2.2 Change of measure and sequential weights

Under standard regularity conditions, Girsanov's theorem gives

$$\begin{aligned} \log \frac{d\mathbb{P}}{d\mathbb{Q}^\phi} = & - \int_0^T b_\tau(\xi_\tau)^\top g(\tau) dW_\tau^\mathbb{Q} \\ & - \frac{1}{2} \int_0^T b_\tau(\xi_\tau)^\top \Sigma(\tau) b_\tau(\xi_\tau) d\tau, \end{aligned} \quad (6)$$

and applying Itô's formula to  $\ell_\tau(\xi_\tau)$  to eliminate the stochastic integral yields the potential/Feynman–Kac representation

$$\log \frac{d\mathbb{P}}{d\mathbb{Q}^\phi} = \ell_0(\xi_0) - \ell_T(\xi_T) + \int_0^T V_\tau(\xi_\tau) d\tau, \quad (7)$$

$$V_\tau(x) = \partial_\tau \ell_\tau(x) + a(x, \tau)^\top b_\tau(x) + \frac{1}{2} b_\tau(x)^\top \Sigma(\tau) b_\tau(x) + \frac{1}{2} \text{tr}(\Sigma(\tau) H_\tau(x)). \quad (8)$$

Discretizing  $[0, T]$  by  $0 = \tau_0 < \dots < \tau_K = T$  with step  $k$  advancing  $\xi_{k-1} \rightarrow \xi_k$ , accumulated diffusion  $\Sigma_k$  and realized increment  $z_k \sim N(0, \Sigma_k)$ , the twist increment telescopes:

$$\Delta f_k := f_{\tau_k}(\xi_k) - f_{\tau_{k-1}}(\xi_{k-1}), \quad f_T(\xi_T) = f_0(\xi_0) + \sum_{k=1}^K \Delta f_k. \quad (9)$$

This gives the common sequential form

$$\log W(\xi_{0:T}) = f_0(\xi_0) + \sum_{k=1}^K G_k + [\log p(y \mid \xi_T) - f_T(\xi_T)], \quad (10)$$

with the *same* exact continuous-time increment represented two ways,

$$G_k^\star = \Delta f_k + C_k = A_k, \quad (11)$$

where the left-endpoint stochastic (Girsanov) increments are

$$C_k = -b_{\tau_{k-1}}(\xi_{k-1})^\top z_k - \frac{1}{2} b_{\tau_{k-1}}(\xi_{k-1})^\top \Sigma_k b_{\tau_{k-1}}(\xi_{k-1}), \quad (12)$$

and  $A_k = \int_{\tau_{k-1}}^{\tau_k} V_\tau(\xi_\tau) d\tau$  is the potential increment. The three standard choices are

$$G_k^{\text{PBS}} = \Delta f_k, \quad G_k^{\text{Girs}} = \Delta f_k + C_k, \quad G_k^{\text{pot}} = A_k, \quad (13)$$

with pseudo-bootstrap (PBS) obtained by dropping the path-measure correction  $C_k$ , and the one-parameter ablation

$$G_k^\lambda = \Delta f_k + \lambda C_k, \quad \lambda \in [0, 1]. \quad (14)$$

The initial weight is  $\log w_0 = f_0(\xi_0)$ ; the terminal correction is  $\log w_K \leftarrow \log w_K + \log p(y \mid \xi_T) - f_T(\xi_T)$ ; and resampling fires when the effective sample size drops below half the particle count (Chopin and Papaspiliopoulos, 2020). Under the exact twist,  $V_\tau \equiv 0$  by the backward Kolmogorov equation, hence  $G_k^{\text{Girs}} = G_k^{\text{pot}} = 0$  and the entire path weight collapses to the boundary factor  $f_0(\xi_0)$  — the zero-variance Doob ideal against which the machinery is validated (Denoising-SMC notes, 2026a,b).

For an Euler–Maruyama proposal the kernel ratio is *exactly* (12) at every step size, because both the unguided kernel  $p(\xi_k \mid \xi_{k-1}) = N(\xi_k; \mu_p, \Sigma_k)$  and the guided kernel  $q(\xi_k \mid \xi_{k-1}) = N(\xi_k; \mu_q, \Sigma_k)$  share covariance  $\Sigma_k$  and differ only in mean, with  $\mu_q - \mu_p = \Sigma_k b_{\tau_{k-1}}$ . Appendix A derives the two cases that matter below. We stress the structural requirement that will reappear in Section 7: *the twist must enter the proposal drift*, i.e. the guided mean shift must be  $\Sigma_k b$ , not  $b$ .

### 3 The DiffusionPDE guided-ODE baseline

#### 3.1 As specified in the paper

The method of Huang et al. (2024) trains one denoiser  $D_\theta$  on the joint (coefficient, solution) field and samples by integrating the probability-flow ODE

$$dx = -\dot{\sigma}(t) \sigma(t) \nabla_x \log p(x; \sigma(t)) dt, \quad \nabla_x \log p(x; \sigma(t)) = \frac{D_\theta(x, \sigma(t)) - x}{\sigma(t)^2}, \quad (15)$$

with guidance approximated à la DPS: under  $y \mid x \sim N(\mathcal{M}(x), \delta^2 I)$ ,

$$\nabla_x \log p(y \mid x_i; \sigma(t_i)) \approx -\frac{1}{\delta^2} \nabla_{x_i} \|y - \mathcal{M}(\hat{x}_N^i)\|_2^2, \quad \hat{x}_N^i = D_\theta(x_i, \sigma(t_i)), \quad (16)$$

so that

$$\nabla_{x_i} \log p(x_i \mid y) \approx \nabla_{x_i} \log p(x_i) - \zeta_{\text{obs}} \nabla_{x_i} \mathcal{L}_{\text{obs}} - \zeta_{\text{pde}} \nabla_{x_i} \mathcal{L}_{\text{pde}}, \quad \zeta = \delta^{-2}, \quad (17)$$

with mean-squared losses

$$\mathcal{L}_{\text{obs}} = \frac{1}{n} \|y_{\text{obs}} - \hat{x}_N^i\|_2^2, \quad \mathcal{L}_{\text{pde}} = \frac{1}{m} \|f(\hat{x}_N^i)\|_2^2. \quad (18)$$

Equation (17) is a *score* (drift) correction.

#### 3.2 As implemented

Algorithm 1 and the code apply the guidance differently. The released implementation in `scripts/generate_burgers.py` (and the siblings `generate_darcy.py`, `generate_poisson.py`, `generate_helmholtz.py`, `generate_ns_*.py`) is, at step  $i$ , with  $\sigma_i := \sigma(t_i)$ :

$$\hat{x}_N = D_\theta(x_i, \sigma_i), \quad d_i = \frac{x_i - \hat{x}_N}{\sigma_i}, \quad (19)$$

$$x_{i+1} = x_i + (\sigma_{i+1} - \sigma_i) d_i, \quad (\text{Heun corrector if } \sigma_{i+1} \neq 0), \quad (20)$$

$$x_{i+1} \leftarrow x_{i+1} - \zeta_{\text{obs}} \nabla_{x_i} \mathcal{L}_{\text{obs}} - \zeta_{\text{pde}} \nabla_{x_i} \mathcal{L}_{\text{pde}}. \quad (21)$$

Three differences from §3 are immediate and all matter.

1. **Placement.** The guidance is applied to the *state* after the ODE step (21), not to the score inside the ODE as in (17). When a drift correction is pulled inside the discretized ODE it acquires the step factor  $(\sigma_{i+1} - \sigma_i)$ ; the flat update does not. We quantify this in Section 4.
2. **Loss normalization.** The code uses unsquared norms,

$$\mathcal{L}_{\text{obs}} = \frac{\|m \odot (u - u_{\text{GT}})\|_2}{128 \cdot 5}, \quad \mathcal{L}_{\text{pde}} = \frac{\|u_t + uu_x - 0.01u_{xx}\|_2}{128 \cdot 128}, \quad (22)$$

with a sensor mask  $m$ , not the squared MSE (18). Hence the implied twist is not Gaussian:  $e^{-\zeta \|\cdot\|_2^2}$  is not a Gaussian likelihood in any noise model.

3. **Scheduling.** The code uses a two-phase rule (observation term only, at full weight, for the first 80% of steps, then observation weight divided by ten plus the PDE term), whereas Algorithm 1 applies both terms at every step. The scheduler is a heuristic anneal of  $\zeta_{\text{obs}}, \zeta_{\text{pde}}$  along the trajectory.

The paper’s own text acknowledges the anneal: the tuned weights “range from 0.1 to 10” and “a significant weight can swiftly decrease the error, but an excessively large weight might cause oversaturation and artifacts” (Huang et al., 2024). This is a reconstruction-oriented tuning, not a likelihood-derived schedule.

## 4 Is the baseline the probability-flow ODE of a twisted marginal?

We now ask the sharp question: does (19)–(21), for some twist  $\phi$ , integrate the PF-ODE that transports the tilted marginal  $\pi_\sigma \propto \phi_\sigma p_\sigma$ ? If so, then in the exact-ODE limit the terminal law would be  $\pi_T \propto \phi_T p_T$ , and the baseline would be a deterministic sampler of that twisted distribution. We show it is not, for any fixed  $\phi$ .

### 4.1 The principled guided PF-ODE

For a variance-exploding model  $x_\sigma = x_0 + \sigma z$ , the heat equation  $\partial_\sigma p_\sigma = \sigma \Delta p_\sigma$  is the continuity equation  $\partial_\sigma p_\sigma + \nabla \cdot (p_\sigma v_\sigma) = 0$  with velocity

$$v_\sigma(x) = -\sigma \nabla_x \log p_\sigma(x). \quad (23)$$

By Tweedie’s identity  $\nabla_x \log p_\sigma = (D_\theta(x, \sigma) - x)/\sigma^2$ , so  $v_\sigma = (x - D_\theta)/\sigma = d_i$ , confirming that (19) is a forward-Euler step of  $v_\sigma$  in the  $\sigma$  variable. For the tilted marginal  $\pi_\sigma \propto \phi_\sigma p_\sigma$  the same continuity argument gives the guided PF-ODE

$$v_\sigma^\pi(x) = -\sigma \nabla_x \log \pi_\sigma(x) = v_\sigma(x) - \sigma b_\sigma(x), \quad b_\sigma = \nabla_x \log \phi_\sigma. \quad (24)$$

Integrating (24) exactly from  $\sigma_{\max}$  down to 0 transports the noise marginal to  $\pi_0 \propto \phi_0 p_0$ ; with the terminally-consistent exact twist  $\phi_0 = p(y \mid \cdot)$  this is the posterior (1). This is the deterministic counterpart of the SMC construction, and it is the idealized object the baseline would have to approximate. (A jitter+ODE variant that makes this exact transport statement robust and connects it to the Feynman–Kac weighting is the subject of Denoising-SMC notes, 2026c.)

### 4.2 What the baseline actually integrates

Take the DPS surrogate  $\phi_\sigma(x) = N(y; \mathcal{M}(D_\theta(x, \sigma)), \delta^2)$  (up to normalization), so that

$$\ell_\sigma(x) = \log \phi_\sigma(x) \simeq -\frac{1}{2\delta^2} \|y - \mathcal{M}(D_\theta(x, \sigma))\|_2^2, \quad b_\sigma(x) = -\frac{1}{2\delta^2} \nabla_x \|\cdot\|_2^2. \quad (25)$$

The principled guided state update from  $\sigma_i$  to  $\sigma_{i+1}$  is obtained by inserting (25) into (24):

$$x_{i+1} = x_i + (\sigma_{i+1} - \sigma_i)(d_i - \sigma_i b_{\sigma_i}(x_i)) = x_i + (\sigma_{i+1} - \sigma_i)d_i - \frac{\Delta\sigma_i \sigma_i}{2\delta^2} \nabla_{x_i} \|\cdot\|_2^2, \quad (26)$$

with  $\Delta\sigma_i := \sigma_i - \sigma_{i+1} > 0$ ; the last term is a *descent* direction in the residual. Thus, if the DPS likelihood is used consistently, its guidance must carry the schedule-dependent coefficient

$$\zeta_{\text{eff}}(\sigma_i) = \frac{\Delta\sigma_i \sigma_i}{2\delta^2}. \quad (27)$$

The baseline (21) instead subtracts a *constant*  $\zeta$  times the residual gradient. Two complementary readings make precise how this differs from the guided PF-ODE.

**Reading 1: intended twist, wrong coefficient.** If  $\zeta$  is intended as the DPS weight (roughly  $1/(2\delta^2)$ ), then the constant replaces the schedule-dependent (27). The ratio

$$\frac{\zeta_{\text{eff}}(\sigma_i)}{\zeta} = \frac{\Delta\sigma_i \sigma_i}{2\delta^2 \zeta} \quad (28)$$

varies by orders of magnitude along a  $\rho$ -spaced schedule and tends to 0 as  $\sigma_i \rightarrow 0$ . Hence no constant  $\zeta$  reproduces the DPS guided PF-ODE at more than one noise level.

**Reading 2: reverse-engineering the target.** One may instead ask which twist would make the flat update a *consistent* guided PF-ODE. The PF guidance term is  $\Delta\sigma_i \sigma_i b$ , so equating it to  $-\zeta \nabla \mathcal{L}$  gives

$$b_{\sigma_i}(x) = -c_i \nabla_x \mathcal{L}(x), \quad c_i := \frac{\zeta}{\Delta\sigma_i \sigma_i}, \quad (29)$$

which is a gradient, namely that of the schedule-tailored twist  $\phi_{\sigma_i}^{\text{base}}(x) = \exp(-c_i \mathcal{L}(x))$ . In other words, the baseline is an Euler step of a guided PF-ODE — but for  $\phi^{\text{base}}$ , not for the DPS likelihood. This yields a sharper statement.

**Proposition 1** (The baseline targets a schedule-dependent, cold reconstruction). *Run for  $K$  steps, the baseline is the Euler discretization of the guided PF-ODE with twist  $\phi_{\sigma_i}^{\text{base}}(x) = \exp(-c_i \mathcal{L}(x))$  and terminal inverse temperature*

$$c_K = \frac{\zeta}{\Delta\sigma_{K-1} \sigma_{K-1}}. \quad (30)$$

*In the exact-flow limit (if the ODE with this twist were integrated exactly) its terminal law is therefore  $\pi^{\text{base}} \propto \exp(-c_K \mathcal{L}(x)) p_0(x)$ , not the posterior (1). Since  $\Delta\sigma_{K-1} \rightarrow 0$  as  $K \rightarrow \infty$  for a fixed schedule endpoint, the target becomes infinitely cold and concentrates on minimizers of  $\mathcal{L}$ . In particular it depends on the number of steps  $K$ : it is not discretization-invariant, and no fixed twist reproduces it across schedules.*

**Remark 1.** *The flat placement is what produces the schedule-tailored twist (29). If the same guidance were instead moved inside the ODE as in (17), its coefficient would be  $\zeta_{\text{eff}}(\sigma_i)$  of (27); the flat update replaces this with  $\zeta$ , which is equivalent to the tailored twist above. The paper is thus internally inconsistent between Eq. (8) (a score/drift correction) and Algorithm 1 (a flat state update); the code implements the latter. The SMC framework’s kernel ratio (Section 7) assumes guidance lives in the drift, i.e. an update of the form  $x_{i+1} = x_i + \Sigma_i(s_{\sigma_i} + b_{\sigma_i}) + z_i$ , which is the step-scaled convention, not the flat one.*

**Remark 2** (Early vs. late weighting). *The mismatch of Reading 1 is not a global rescaling: relative to (27) the baseline under-weights guidance at large  $\sigma$  and over-weights it at small  $\sigma$ , where the required coefficient vanishes but the constant does not. This is exactly the signature of the effective inverse temperature  $c_i$  of (29) growing without bound as the trajectory denoises, which is why the baseline behaves as a penalized reconstruction rather than a posterior sampler.*

## 5 Structural comparison

Table 1 summarizes the two constructions. The rows are chosen so that every difference is a property of the algorithm, not of implementation detail.

Table 1: The Girsanov-corrected SMC framework versus the DiffusionPDE guided-ODE baseline.

|                           | <b>Girsanov-corrected SMC</b><br>(Denoising-SMC notes, 2026a)                                        | <b>DiffusionPDE baseline</b> (Huang et al., 2024)                                                                                                                                   |
|---------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target                    | Posterior $p(x_0   y) \propto p(y   x_0)p(x_0)$ ; exact up to discretization and terminal correction | Implicit target $\propto \exp(-c_K \mathcal{L})p_0$ with $c_K = \zeta/(\Delta\sigma_{K-1}\sigma_{K-1}) \rightarrow \infty$ ; a schedule-dependent reconstruction, not the posterior |
| Dynamics                  | Stochastic reverse SDE (4)                                                                           | Deterministic PF-ODE (15), Heun                                                                                                                                                     |
| Guidance placement        | Drift: $x_{i+1} = x_i + \Sigma_i(s + b) + z_i$                                                       | Flat state update after the step, with or without a constant schedule                                                                                                               |
| Guidance scaling          | $\Sigma_i b = \delta_i b$ (step-scaled)                                                              | $-\zeta \nabla \mathcal{L}$ , $\zeta$ constant                                                                                                                                      |
| Guidance magnitude source | Twist gradient $b = \nabla \log \phi$ , with the PF coefficient $\sigma$                             | Tuned $\zeta$ ; loss is an unsquared residual norm                                                                                                                                  |
| Inference object          | Weighted particle population; approximates a distribution                                            | A single deterministic field per noise draw                                                                                                                                         |
| Weights / resampling      | Girsanov/PBS/potential weights; systematic resampling at $\text{ESS} < N/2$                          | None                                                                                                                                                                                |
| Uncertainty               | Posterior spread, ESS available                                                                      | None                                                                                                                                                                                |
| Consistency               | Yes, for the posterior, as $K, N \rightarrow \infty$ under regularity                                | None claimed; not derived from a change of measure                                                                                                                                  |
| Metric                    | Posterior functionals (mean, credible region, calibration)                                           | Relative $L^2$ to one ground-truth field                                                                                                                                            |

## 6 Three distinct objects, not two

A frequent source of confusion is that this repository contains a third, intermediate update, the deterministic limit of the SMC proposal (the `--no-noise --rho 0` control in `scripts/generate_burgers_gem.py`). With no injected noise and unweighted particles, the GEM step is

$$x_{i+1} = x_i + \delta_i(s_{\sigma_i}(x_i) + b_{\sigma_i}(x_i)), \quad \delta_i = \sigma_i^2 - \sigma_{i+1}^2, \quad s_{\sigma_i} = \frac{D_\theta(x_i, \sigma_i) - x_i}{\sigma_i^2}. \quad (31)$$

This is *not* the baseline. With  $b \equiv 0$ , the ratio of the unguided parts of (31) and (19) is

$$\frac{\delta_i s_{\sigma_i}}{(\sigma_{i+1} - \sigma_i) d_i} = \frac{\frac{\delta_i(D_\theta - x_i)}{\sigma_i^2}}{\frac{(\sigma_{i+1} - \sigma_i)(x_i - D_\theta)}{\sigma_i}} = \frac{\sigma_i + \sigma_{i+1}}{\sigma_i} \in (1, 2], \quad (32)$$

using  $\delta_i = (\sigma_i - \sigma_{i+1})(\sigma_i + \sigma_{i+1})$ . So the deterministic SMC limit takes a systematically larger score step (up to a factor 2 at small  $\sigma$ ) than the baseline’s PF-Euler step, even before guidance. The correct comparison is therefore three-way:

- (A) the baseline: PF-ODE Heun + flat loss-gradient, constant  $\zeta$ ;
- (B) the guided SDE’s deterministic limit:  $\delta_i$ -scaled score and guidance, no noise, no weights (the natural same-target ablation);
- (C) the full stochastic GEM + Girsanov SMC.

(A) and (C) differ in both target and dynamics; (B) and (C) differ only in the stochasticity and the reweighting, and are the pair that can be compared as two estimates of the same posterior.

Using (A) as the reference against which (C)’s relative error is judged conflates target mismatch with sampling quality.

## 7 Two implementation-level caveats in the SMC code

The preceding sections concern the baseline. The next two observations concern the SMC code itself, where the baseline convention or a non-likelihood residual has been imported into places the theory does not cover. Both are reported here, not fixed.

### 7.1 Flat guidance invalidates the closed-form kernel ratio

The identity (12) is the exact Gaussian kernel log-ratio  $\log \frac{p(\xi_k|\xi_{k-1})}{q(\xi_k|\xi_{k-1})}$  only when the guidance enters the drift. Appendix A derives both cases:

$$\text{drift-embedded } (\mu_q - \mu_p = \Sigma_k b) : \quad C_k = -b^\top z_k - \frac{1}{2} b^\top \Sigma_k b = -b^\top z_k - \frac{1}{2} \delta_k \|b\|^2, \quad (33)$$

$$\text{flat } (\mu_q - \mu_p = b) : \quad \tilde{C}_k = -\frac{1}{\delta_k} b^\top z_k - \frac{1}{2\delta_k} \|b\|^2, \quad (34)$$

in the scalar-covariance case  $\Sigma_k = \delta_k I$ . These agree only if  $\delta_k = 1$ , which is not the case for any realistic step. Since  $\delta_k$  varies across the schedule, (34) and (33) cannot be reconciled by any constant rescaling of the weight either. Consequently the `--flat-guidance` control of `scripts/generate_burgers_gem.py`, which switches the proposal to  $x_{i+1} = x_i + \delta_i s_{\sigma_i} + b_{\sigma_i} + z_i$  while still calling `girsanov_increment` (i.e. (33)), attaches a correction that is not the kernel ratio of the proposal actually used. The self-check `smc/scripts.2/check_gem_tds_real_model.py` validates (33) on the real network and therefore applies only to `scale_guidance=True` (the default); it does not certify the flat mode. Any experiment that seeks to compare against the baseline convention must first derive the flat kernel ratio (34) (or, equivalently, stop calling it a Girsanov weight).

### 7.2 The PDE residual is not a likelihood

The SMC target (1) presumes a specified observation likelihood  $p(y \mid x_0)$ . The Burgers runner instead defines the twist from the unsquared losses (22),

$$\ell_\sigma(x) = -\zeta_{\text{obs}} \|m \odot (u - u_{\text{GT}})\|_2 / (128 \cdot 5) - \zeta_{\text{pde}} \|u_t + uu_x - \nu u_{xx}\|_2 / (128 \cdot 128), \quad (35)$$

so that the effective target is a *tempered surrogate*  $p(x_0) \exp(-\zeta_{\text{obs}} \|\cdot\|_2 - \zeta_{\text{pde}} \|\cdot\|_2)$  rather than  $p(x_0)p(y \mid x_0)$ . Two consequences follow.

First, the terminal correction of (10) is formally required for exactness whenever  $\phi_0 \neq p(y \mid \cdot)$ . The runner omits it and instead asserts terminal consistency “by construction”, on the grounds that the same loss is the only notion of fit available at every step including the last (`scripts/generate_burgers_gem.py`, comments around the boundary-weight initialization). That is a modeling choice, not a theorem: it fixes the target to the surrogate (35), not to the Bayesian posterior of any stated observation model. If a genuine likelihood is intended, either  $L(x_0)$  must be specified and the terminal correction retained, or the surrogate must be shown terminally consistent relative to that  $L$ .

Second, because the residual is included in  $\ell$  with a fixed  $\zeta_{\text{pde}}$ , the “PDE constraint” is a soft prior whose weight is not derived from a noise model. Combined with the two-phase schedule, the effective target drifts over the trajectory. This is defensible as a regularized posterior, but it should

be stated as such, and it means the SMC relative-error metric does not measure fidelity to the Bayesian posterior the `note_1` framework is built to estimate.

## 8 Implications for experiments

The analysis suggests the following comparison protocol, which separates the three objects of Section 6 and the two notions of “error”.

1. **Choose a likelihood.** Specify  $p(y | x_0)$  explicitly (e.g. a Gaussian sensor model with stated  $\delta$ ) and use it as  $L$ ; keep the PDE residual either out of the twist or explicitly declared as a prior. Then the target is unambiguous.
2. **Retain the terminal correction** (or prove terminal consistency), so the sampler targets the stated posterior rather than a surrogate.
3. **Do not mix conventions.** If flat guidance is desired, use (34); if the theory’s weight is desired, use drift-embedded guidance.
4. **Compare like with like.** Contrast (C) against (B), its own deterministic and weighted-off limit, to isolate the value of stochasticity and reweighting. Treat (A) as a separate reconstruction method, not as the same target.
5. **Report target-appropriate metrics.** For (B) and (C), posterior functionals (weighted mean, spread, calibration, ESS); relative  $L^2$  to a single ground-truth realization is a reconstruction metric and can be minimized by point estimators that need not be posterior samples.

## 9 Summary

Table 2 maps the repository artifacts to the methods and targets discussed here.

| Table 2: Repository artifacts, methods, and targets.    |                                    |                                                |                                                                           |
|---------------------------------------------------------|------------------------------------|------------------------------------------------|---------------------------------------------------------------------------|
| Artifact                                                | Method                             | Update                                         | Target                                                                    |
| <code>scripts/generate_*.py</code>                      | DiffusionPDE baseline (A)          | PF-ODE Heun + flat $-\zeta \nabla \mathcal{L}$ | schedule-dependent cold reconstruction (Proposition 1); not the posterior |
| <code>generate_burgers_gem.py --no-noise --rho 0</code> | Guided-SDE deterministic limit (B) | $x + \delta s + \delta b$                      | twisted surrogate (same as C, zero-variance)                              |
| <code>generate_burgers_gem.py</code> (default)          | GEM + Girsanov SMC (C)             | $x + \delta(s + b) + z$ , weighted             | posterior (up to discretization, terminal correction)                     |
| <code>scripts_2/weightings/doob_vtau.py</code>          | Potential/Feynman–Kac weighting    | $A_k = \int V_\tau d\tau$                      | same as (C), Hessian-trace form                                           |

The central conclusion is a scope statement: the Girsanov-corrected SMC construction estimates the posterior (1); the DiffusionPDE baseline does not define or estimate that object. They are two different things that happen to share a denoiser and a sparse-observation loss, and they should not be placed on the same target axis. Within the SMC construction, the flat-guidance control and

the use of an unnormalized PDE residual as the twist are the two places where the implementation currently leaves the theory, and both are documented above rather than changed here.

## A Gaussian kernel ratios for drift-embedded and flat guidance

Consider Gaussian kernels with common covariance  $\Sigma_k$ ,

$$p(\xi_k | \xi_{k-1}) = N(\xi_k; \mu_p, \Sigma_k), \quad q(\xi_k | \xi_{k-1}) = N(\xi_k; \mu_q, \Sigma_k), \quad \xi_k = \mu_q + \Sigma_k^{1/2} \varepsilon_k, \quad (36)$$

so the realized increment is  $z_k = \Sigma_k^{1/2} \varepsilon_k$ . Writing  $\|\cdot\|_{\Sigma_k^{-1}}^2$  for the  $\Sigma_k^{-1}$ -norm,

$$\log \frac{p}{q} = -\frac{1}{2} \left( \|\xi_k - \mu_p\|_{\Sigma_k^{-1}}^2 - \|\xi_k - \mu_q\|_{\Sigma_k^{-1}}^2 \right). \quad (37)$$

**Drift-embedded guidance.** If  $\mu_q - \mu_p = \Sigma_k b$ , then  $\xi_k - \mu_p = \Sigma_k b + z_k$  and  $\xi_k - \mu_q = z_k$ , so

$$\log \frac{p}{q} = -\frac{1}{2} \left( \|\Sigma_k b + z_k\|_{\Sigma_k^{-1}}^2 - \|z_k\|_{\Sigma_k^{-1}}^2 \right) = -b^\top z_k - \frac{1}{2} b^\top \Sigma_k b, \quad (38)$$

which is (12) and, for  $\Sigma_k = \delta_k I$ , (33).

**Flat guidance.** If instead  $\mu_q - \mu_p = b$ , then  $\xi_k - \mu_p = b + z_k$  and

$$\log \frac{p}{q} = -\frac{1}{2} \left( \|b + z_k\|_{\Sigma_k^{-1}}^2 - \|z_k\|_{\Sigma_k^{-1}}^2 \right) = -b^\top \Sigma_k^{-1} z_k - \frac{1}{2} b^\top \Sigma_k^{-1} b, \quad (39)$$

which is (34) for  $\Sigma_k = \delta_k I$ . The two expressions coincide only at  $\Sigma_k = I$ ; they cannot be related by a constant rescaling when  $\Sigma_k$  varies along the schedule.

## B ODE coefficient comparison

Let  $d_i = (x_i - D_\theta(x_i, \sigma_i))/\sigma_i = v_{\sigma_i}(x_i)$  and  $s_{\sigma_i} = (D_\theta(x_i, \sigma_i) - x_i)/\sigma_i^2 = -d_i/\sigma_i$ . Then

$$(A) \text{ unguided: } (\sigma_{i+1} - \sigma_i)d_i, \quad (B) \text{ unguided: } \delta_i s_{\sigma_i} = -\frac{\delta_i}{\sigma_i} d_i, \quad (40)$$

and the ratio is

$$\frac{(B)}{(A)} = \frac{-\delta_i/\sigma_i}{\sigma_{i+1} - \sigma_i} = \frac{\delta_i}{\sigma_i(\sigma_i - \sigma_{i+1})} = \frac{\sigma_i + \sigma_{i+1}}{\sigma_i}, \quad (41)$$

using  $\delta_i = (\sigma_i - \sigma_{i+1})(\sigma_i + \sigma_{i+1})$ . The two deterministic recursions therefore differ by a factor in  $(1, 2]$  even with no guidance at all.

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